Pre-Thesis I Report

An Implementation and Analysis of Deep Learning Models for the Detection of Wheat Diseases

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Declaration

It is hereby declared that

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3. The report does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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Abstract

Detecting a disease visually is a time-consuming and error-prone operation, and in the agricultural arena, for disease control, crop yield loss prediction, and global food security, automatic and accurate evaluation of disease severity in crops is a particularly demanding study area. Deep Learning (DL), the latest innovation in the era of Artificial Intelligence (AI), is promising for fine-grained categorization of crop diseases since it eliminates labor-intensive feature extraction and segmentation. To diagnose the disease from photos, we intend to construct a deep learning model based on Convolutional Neural Network (CNN) architecture. In addition, we want to develop an mobile app that would allow the farmers to identify the disease in real-time using their cameras and receive instructions to take immediate action to stop the disease from spreading.

Keywords: Deep Learning; Machine Learning; Wheat; Diseases; Prediction; Hyperspectral; Imaging; Data augmentation; Image segmentation

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Nomenclature

The next list describes several symbols & abbreviation that will be later used within the body of the document

CNN : Convolutional Neural Network

$DCNN$: Deep Convolutional Neural Network

DL : Deep Learning

FHB : Fusarium head blight

$k - NN$: K-Nearest Neighbour

KNN : K-Nearest Neighbour

ML : Machine Learning

NB : Naive Bayes

NIR : Near-infrared

NN : Neural Network

$RCNN$: Region-Based Convolutional Neural Network

RF : Random Forest

RGB : Red-Green-Blue

SVM : Support Vector Machine

Chapter 1

Introduction

1.1 Importance of Wheat

After Bangladesh got independence in 1971, rice was the main crop that people used to eat. During the period from 1971 to 1975, the rice price was increasing globally which caused food shortages in Bangladesh. In addition, Bangladesh was already lagging behind in food production due to the War of Liberation along with natural disasters such as floods, droughts, cyclones and population growth. It was then decided rice alone could not fulfill the nation's food demand. As a result, wheat was chosen as an alternative winter crop. Since then wheat has been growing in popularity in Bangladesh but climate change has a significant impact on Bangladesh and so the likelihood of growing wheat and potatoes would be substantially hampered by a shift in the average temperature of 2-4 °C, and output loss may surpass 60% of the possible yields [1].
According to [27] wheat consumption in 2021 has seen twice as much demand as in 2015. The reasons behind this are the good quality flour being cheaper than medium quality rice, the increase of the obese and diabetic patients, and the rise in the popularity of bakery products such as biscuits and cakes. Thus the imports have seen a 116% rise in the past 6 years. [11] However, the production of wheat hasn't been able to keep up with the demand of the ever growing population.
On the other hand, [26] global food demand has been increasing from COVID-19 and the Russia-Ukraine war has further increased that. Almost 90% of wheat and other grains are shipped from Ukraine, the "breadbasket of the world" but due to the war, the crops could not leave Ukraine. Thus the global wheat demand is increasing. In addition, the crop fields that were destroyed during the war will not get us food in the next harvesting season which will increase the global food shortage even more. As India is one of the countries Bangladesh imports wheat from, the restriction on wheat export from India escalates the wheat demand problem for Bangladesh. Lastly, there are several wheat diseases and most of them have been dealt with in various cultivars. However, [13] Bangladesh witnessed the Wheat Blast disease for the first time in February of 2016 which was caused by MoT. Wheat production decreased to 0.349 million acres in the 2017–2018 growing season, which was the lowest level in the previous three decades. As a result, the Bangladeshi government advised farmers to abandon wheat in favor of other crops including lentil, boro rice, maize, etc. and this led to a 52% decrease in the areas used for wheat growing. All these reasons are causing a reduction in wheat production and farmers are afraid to

cultivate wheat. This is why it is really critical to detect diseases at an early stage or even correctly so that the little amount that's being cultivated can be harvested in full.

1.2 Some CNN Models

Deep learning algorithms have been used extensively in research up to this point. Among these, [7] NN has been used along with support vector machines (SVM), SVM, and NN achieving an accuracy of 89.23% and 80.21% respectively. [21] An enhanced VGG16 model that outperformed the single-task learning approach was suggested. [9] Then, a CNN was employed to categorize the three wheat illnesses of Powdery mildew, Yellow rust, and Stem rust. [12] A pre-trained CNN using GoogleNet architecture was used where the overall accuracy was 94%. [20] Another proposed model got a testing accuracy of 97.88%, representing improvements of 7.01% and 15.92% over VGG16 and RESNET50, respectively. [6] Further, between AlexNet and GoogleNet, GoogleNet performed better, having an accuracy of 99.35%. [15] Another model that was evaluated using images from the Sukkur IBA University performed with an accuracy of 98% and was trained using 38 classes from the PlantVillage dataset. [8] A different image processing method used crucial programs such as GLCM, PNN, and k-mean clustering. [4] The VGG-FCNVD16 model trained on the WDD2017 database showed an accuracy of 98.84%. [22] Improved Inception-v3 and ResNet-152 have a corn crop accuracy of 97.81% and 97.48%, respectively. [14] Another study used DL and image processing techniques to precisely identify ill regions in color photos of wheat spikes. To retrain a Mask RCNN model learning technique to recognize and distinguish spikes in images, many labeled photos were divided into training and validation datasets. The sick areas of the spikes in the grayscale images were suggested to be highlighted using a novel color feature called GB. The enhanced region-growing algorithm successfully identified the FHB-infected regions of each spike. The results showed that the RG algorithm outperformed K-means and Otsu's method for segmenting ill regions.[17] Furthermore, a more generalized model based on hyperspectral imaging was created by comparing the spectra and images from FHB-infected wheat heads using Random Forest, Partial least squares regression, Support Vector Regression, and CNN. [10] Another hybrid model has the best generalization performance for F1 score and accuracy, which are 0.75 and 0.743, respectively while using a more extensive dataset for testing.

There are pre-trained models such as VGG16, AlexNet and ResNet50 which have been trained on a different dataset but performing really well in the case of a different dataset. Thus, our proposed CNN model will be detecting all the wheat diseases there are as no one else has done this before as far as our research goes. This will be beneficial to our farmers for detecting [5] diseases with the same symptoms, multiple diseases at the same time or contagious diseases like a Wheat Blast from spreading and reduce the loss as it will be faster than traditional laboratory test methods and less labor-intensive.

1.3 Problem Statement

There have been many research works until now, where there are some works with SVM, VGG16, GoogleNet, ResNet50, GLCM,PNN and many CNN models too. Alongside, few of the authors tried to get a higher percentage in the accuracy rate using some hybrid models. Some of the models did fascinating work for a few wheat disease detection with great accuracy. However, most of the research conducted till now had a limited number of images for training the models and those models could be more efficient and robust by using a larger dataset. Again, some deep CNN models need so much resources to run which could be improved in the future by fine tuning parameters or by reducing the number of fully connected layers. There are papers with wheat diseases but there have not been any research on the diseases with the same symptoms or all the known diseases of wheat.

We intend to work on a private dataset that incorporates all known wheat diseases. It will also contain diseases that share similar symptoms among them. We will pay particular attention to the Wheat Blast, which, if not discovered quickly, might harm the entire field. In addition, we will include more photographs Wheat Blast, which, as far as we are aware, has never been done before. Our custom dataset will be fed to our proposed CNN model and then it will be implemented in an android application. Therefore, the users can detect all kinds of wheat diseases much faster and with more ease. Furthermore, our system will also provide recommendations based on the severity of the detected disease(s) so that they can take wise decision as to whether to prune the plants or spray pesticides and other medicines. In order to provide farmers with a user-friendly experience, this app will also support Bangla language.

1.4 Research Objectives

With the use of CNN, we hope to create a deep-learning model that can identify various wheat diseases such as Wheat Blast, Leaf Rust, Stem Rust, Glume Blotch, Common bunt, Stripe bust,Wheat streak mosaic, Tan spot etc. The user will take a picture of a leaf which seems to be infected to our system. The image will be run through our proposed CNN model to check for the presence of an infected leaf after completing all necessary preprocessing. The farmers will be given possible disease(s) and the actions they can take depending on the severity of the disease(s). There have been several CNN models as well as some hybrid ones that have performed really well but studies have not been done on the wheat diseases of the same symptoms e.g. FHB and Wheat Blast.

After conducting this study, we look forward to:

1. Have a solid understanding of the Deep Learning Model and AI.
2. Develop a model that will detect whether a wheat leaf is diseased or not so that wheat cultivation can be benefitted.
3. Compare the CNN model with other models.
4. Make an android application that will be convenient for our farmers.

Chapter 2

Background

2.1 Literature Review

[7] The authors have collected the images first, then they preprocessed the images, after that they did the image segmentation process and lastly they did feature extraction which included color, shape and size. They have used a Neural Network (NN) and Support Vector Machine (SVM) for classification and SVM got 89.23% accuracy while NN got 80.21%. [2] There isn't a unified image segmentation method for some of the prevalent wheat diseases because the diseases that affect wheat leaves are diverse. The authors tried to solve it by using the K-means clustering algorithm for an automatic and efficient segmentation of color images because coloured photographs contain more information than grayscale ones. Wheat leaf diseased areas are automatically distinguished from healthy ones using the K-means clustering technique and then the image is translated from RGB to Lab color space, and the pixels in the latter are clustered according to how close the pixels are to one another. This method increases the accuracy to over 90% for 3 well-known diseases of wheat: stripe rust, leaf rust and powdery mildew.

[18] The authors have used an image hashing algorithm to create a dataset of 2414 images where 80% were labeled as single-diseased, more than 12% were labeled as healthy plants and around 6% were labeled as multiple-diseased. The method utilized for disease recognition is based on the EfficientNet-B0 neural network architecture. Among the 3 techniques they applied, the first one was transfer learning, the second one was grouping data for training, validation and testing samples, the last one involved the transfer and augmentation of image styles. Among these, transfer learning and augmentation performed with accuracy of 0.942 which was better than the second one.

[19] They have classified the severity of wheat blast into 3 categories using DCNN. They trained their model on 2 different datasets one containing both premature and maturing spikes and the other containing only the premature spikes. When identifying the photos, the models trained on only premature wheat spikes displayed greater recall, precision, and F1 score than the models trained on both types. When a wheat spike matures the color changes to yellow/white which can confuse the CNN model. Again, FHB may confuse the model as well as their symptoms are quite similar (e.g. spike bleaching). This study has been conducted only on wheat blast and

so the CNN model could be made more generalized by using a larger dataset and variety of diseases. [21] The improved VGG16 model, trained on a small dataset, performs better than some methods including single-task learning, ResNet50, the reuse-model of transfer learning and DenseNet121. This multi-task transfer learning for recognizing the diseases of wheat leaf and rice is better in generalizing while it is able to reduce the number of parameters.

[9] CNN has been used for classifying three wheat diseases which are Stem rust, Yellow rust, Powdery mildew. Their dataset contained four classes, three of which were the diseases mentioned earlier and the other one was healthy leaves. Each of the classes contained 2707 images. AlexNet was used for training the dataset. [12] We are not getting the full potential of Deep learning techniques due to small numbers and less diversity of images. This is why they proposed Data augmentation to increase the number of images and to increase the diversity they applied image segmentation to work with lesions and spots instead of the whole image which helped them to get multiple conditions from a single image. They used a pre-trained CNN using GoogleNet architecture where the overall accuracy they got is 94% by using individual lesions and spots higher than the ones using original images. However, the model could be more accurate if there were more data.

[20] The accuracy of a deep learning model varies depending on the size, the class imbalance and the variety of the dataset and so the authors have worked with not only leaves but also spikes and roots of a wheat plant in this paper. They have compared their proposed model against pre trained deep learning architectures ResNeT50 and VGG16. All the models were trained up to 1000 epochs and the proposed model has been trained with over 12,000 images of wheat diseases. The model has classified wheat diseases of 10 classes of the LWDCD2020 dataset with an training accuracy of 98.62% and a testing accuracy of 97.88% which is 7.01% and 15.92% improvement against VGG16 and ResNet50 respectively. [25] Here the author proposed a model that combines a logistic regression mechanism-based and KNN to address the low accuracy and poor stability for the prediction of wheat FHB and developed a remote sensing approach of predicting FHB. The KNN model was used to predict the prevalence of FHB using the disease prediction variables along with factor weights. The outcomes show that the logistic-KNN model can increase the stability and predictability of machine learning models without considering wind speed predictors. There will be many computations for the KNN model. In future, they may implement k-d tree to simplify model's computations and increase the model's applicability.

[6] Their goal was to classify crop species, presence of any disease and identity of the disease if there was any. The dataset for classification was collected from the PlantVillage project and it contained 54,306 images among which the number of crop species and crop diseases were 14 and 26 respectively. Among the two compared CNN architectures which are AlexNet and GoogleNet, the latter performed better both in terms of transfer learning and training from scratch. 993 images out of 1000 were correct when their model tried to classify crops and diseases resulting in an accuracy of 99.35%. However, if the model was to identify any images that were taken under different conditions than the ones that have been used for training

the accuracy drops to 31%. The accuracy of the model can be improved by creating a more diverse dataset for training where leaves may face downwards or diseases may occur at different parts of the plant. [15] As detecting plant diseases by looking at leaves, stems or fruits is a job requiring human resources, the authors have used CNN to detect plant diseases from the PlantVillage dataset where they worked with 15% of the total images. The model trained using 38 classes of the dataset was tested using the images of Sukkur IBA University and it performed with an accuracy of 98%. There could be more actual environmental images, plant types and diseases for improving the accuracy of the model. In future the model may adopt a 3 layer approach where the first layer will be detecting plants in an image, the second layer will be telling the plant type and the last layer will be used for classifying diseases.

[8] The paper is written based on India which is an agricultural country as well as Bangladesh. Farmers are not aware of what types of diseases plants have and how to prevent these. Utilizing an image processing method to find plant diseases is one solution. Image acquisition, pre-processing of images, extraction of features, and finally, a classifier known as a neural network are included in this procedure. In order to identify leaf disease, a typical image is first captured using any digital camera, and the image quality is then improved through pre-processing. Afterward, the image is split into clusters with the infected area remaining and the green pixels being masked. Segmentation is the technique used after the masking to increase precision. Feature extraction and statistical analysis are completed following segmentation. A neural network is finally used to classify the images. The image processing method used for this plant disease identification makes use of crucial programs including GLCM, PNN, and k-mean clustering. One of the difficulties is that the plant needs to be regularly watched in order to discover the illness in its early stages. In future work, a large number of data must be used for training purposes in a neural network. The increment of training data results in the accuracy of the system being high. And fuzzy logic can be used as a classification tool and the accuracy rate and speed of the system can be compared.

[24] This work demonstrates the detection and classification of diseases that damage wheat harvests using a two-dimensional CNN model. The report mentioned the use of VGG-FCNVD16 as a method to circumvent the issues and achieve an average production of 97.95 percent. They used WDD-2017 as the dataset. Similar image pre-processing on the acquired dataset, importing the required elements, opening the dataset and loading the photos, the ability to export binary images as pixel arrays after conversion, etc. After gathering 4800 photos in total for this study, the dependability of the results was determined to be 98.84 percent. designing software that can manage more types of diseases by adding new disease classes to its capability list and using multi-dimensional CNN could increase the accuracy of disease detection.

[4] In this study, they used near-infrared (NIR) hyperspectral imaging for extracting the contaminants from wheat. Then, raw NIR reflectance spectra of these pollutants and wheat were collected (in a range of 1000–1600 nm in NIR). The raw and pre-processed data were categorized using SVM, Naive Bayes (NB), and k-nearest neighbors (k-NN) classifiers. To comprehend the categorization of each form of

wheat contamination as well as the classification of all wheat contaminant types, two-way and multi-way classifications were applied in each study. Contaminants and wheat had the highest classification accuracy when spectral data were pre-processed using the SNV method and categorized using a k-NN classifier. After applying different approaches the authors got the results that showed that canola was most accurately classified as a foreign material, while oats were the least accurately classified. The three classifiers (SVM, NB, and k-NN) all had accuracies of around 79.0±0.8%. [22] The study suggested using ResNet-152 and Inception-v3 versions to identify diseases in essential crops like rice and corn and they both added layers to those models to improve accuracy. Incorporating a new dense layer into both models improved overall disease classification and identification accuracy using the dataset of rice and maize leaves. For the corn crop, the suggested method's accuracy was 97.81% and 97.48%, respectively, using variants of InceptionV3 and ResNet152. More deep learning models, including NesNetMobile and NesNetLarge, as well as AlexNet, VGG and all of its variants, Xception, MobileNet, and DenseNet, are required to examine accuracy in this work. An expanded version of this research work can be used to investigate additional plant leaf diseases using the same methods.

[14] This paper demonstrates how to detect FHB disease in wheat crops using DL and image processing methods. Color images of wheat spikes taken at the milk stage were processed to create datasets, which is used for developing a reliable and cost-effective high-throughput phenotyping system (helpful for developing resistant wheat varieties). A Mask R-CNN model was retrained through the transfer learning technique to successfully detect and identify spikes in images. Only the last few layers of the network were trained, resulting in an accurate detection rate for most spikes across several wheat varieties with different morphologies and sizes. R2, RMSE, and rRMSE all remained at 0.80%, 1.17%, and 21.37% respectively when assessing the model's accuracy using testing data sets compared to manual counts alone. The mAP achieved was 0.9201. The sick areas of the spikes in the grayscale images were suggested to be highlighted using a novel color feature called GB. The FHB-infected regions of each spike were successfully detected by the improved region-growing algorithm. The segmentation of sick regions using the RG algorithm outperformed using K-means and Otsu's technique, according to the results. It has been shown that DL algorithms enable precise FHB detection in wheat based on color picture analysis, increasing the effectiveness of FHB assessment in the field. The results of testing 450 images showed that it is possible to detect FHB in wheat using color images.

[17] The spectral and image of wheat spikes infected with FHB have been collected for comparing Support Vector Regression, Random Forest (RF), Partial least square regression and CNN for developing a more generalized model based on hyperspectral imaging. The deep CNN (DCNN) model compared better than others as it scored 0.97 and 3.78 in R^2 and RMSE respectively. However, DCNN needs high hardware requirements and GPU support and more studies could be conducted to examine the impacts of different locations, years and wheat cultivars on the performance of the model since these factors may affect hyperspectral imaging and make the model more robust. [10] The authors tried to detect FHB in early stages using hyperspectral imagery. Instead of traditional methods they have used a 2D data structure as

the input for their CNN and then used a bidirectional recurrent and a convolutional layer for constructing a hybrid NN which would generalize the CNN model. The mode performed better than other deep learning models as it scored 0.75 and 0.743 in F1 score and accuracy respectively.

[3] Improving the detection of mycotoxin deoxynivalenol (DON), formed by FHB, in wheat kernels due to the health hazards connected with its consumption is a primary research goal. This study contrasted the speed of human work to the quicker process and HSI is used as the basis for more reliable detection techniques. Here, two methods are used for FHB detection:the first uses images of wheat ears captured in the field, while the second uses images of kernels normally taken in a lab under controlled circumstances. First off, the created technique proved resistant to issues including form, orientation, shadowing, and clustering of kernels, with a classification accuracy of 91%. In addition to FHB detection, it was shown that the system could calculate DON concentrations in wheat kernels, albeit to a limited extent. Additionally, Delwiche, Kim, and Dong presented a study on the evaluation of Fusarium damage in wheat kernels using NIR and VR images in 2011 and with these, they used linear discriminant analysis (LDA) and obtained an accuracy of 95%.

[16] In order to analyze the spectroscopic diagnosis of FHB, this study used continuous wavelet analysis (CWA) to examine the reflectance spectra of wheat ears (350 to 2500 nm). Hyper-spectral remote sensing has assisted in the identification of FHB in wheat ears. Additionally, employing CNN in a natural field, healthy and FHB-infected wheat could be distinguished. It's been noted that CWA did a good job at identifying crop stressors,monitoring diseases and pests. During the stages of wheat filling in 2018 and 2019, two separate hyper-spectral datasets in range of 350-2500nm were used. The objectives of this work were to: (1) evaluate CWA's performance; and (2) determine the optimal wavelet features and both to detect FHB. An FLDA model was constructed using the six wavelet qualities as a base. The wavelet characteristics performed well in diagnosing FHB in winter wheat ears, as evidenced by an accuracy around 88.7% and a 0.775 kappa coefficient. The identified future plan included testing the practicality of CWA combining different spectral acquisition angles to find FHB in winter wheat at field scales using UAV hyper-spectral technology.

[23] Here the author tried to identify wheat seeds from the TBIO Toruk cultivar that were FHB-infected using various deep-learning techniques based on RGB photos, integrating hyperparameter optimization and fine-tuning techniques with various pretrained CNN.The models detected FHB in seeds with a 99% accuracy and a 97% accuracy utilizing a low-complexity design architecture with hyperparameter optimization. These results point to the potential for accurate categorization of wheat seeds with FHB using low-cost imaging equipment and deep-learning models. Convnets were used to distinguish between sick seeds and healthy seeds using data collected from RGB digital photos in order to build deep-learning models for monitoring FHB in wheat seeds.The behavioral changes among wheat genotypes may affect the detection method's accuracy because FHB symptoms are genotype dependent.

Chapter 3

Work Plan

Firstly, we will manage a private dataset from from an institute like Bangladesh Agricultural University or any institution dedicated to agriculture. The data will next be pre-processed using a variety of approaches, including data splitting, data augmentation, and data resizing.Then, utilizing the training dataset and various methodologies, we will construct a CNN model, train it using the pre-processed images and then evaluate accuracy using the testing dataset with the help of accuracy, precision and some other techniques. We must determine whether the precision is adequate because there may be some trial and error. If yes, we can proceed with designing an Android application for users to easily learn the condition of wheat illness and save this model.

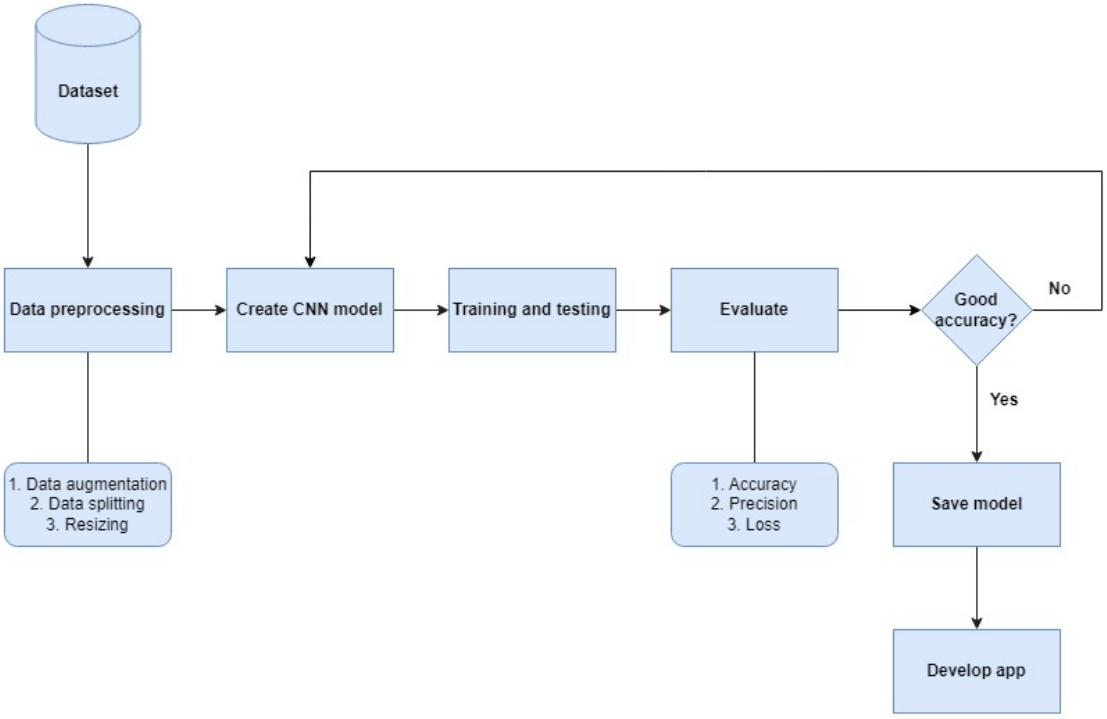


Figure 3.1: Work Plan

The software will prompt the user to capture pictures of wheat plants while they are being used. Later, these photos will be pre-processed using a few methods and fed into the CNN model we've suggested. The user will then see several notifications based on the result of the model, including suggestions for immediate action to stop the disease from spreading further and assurances to the farmers over the health of their wheat plants.

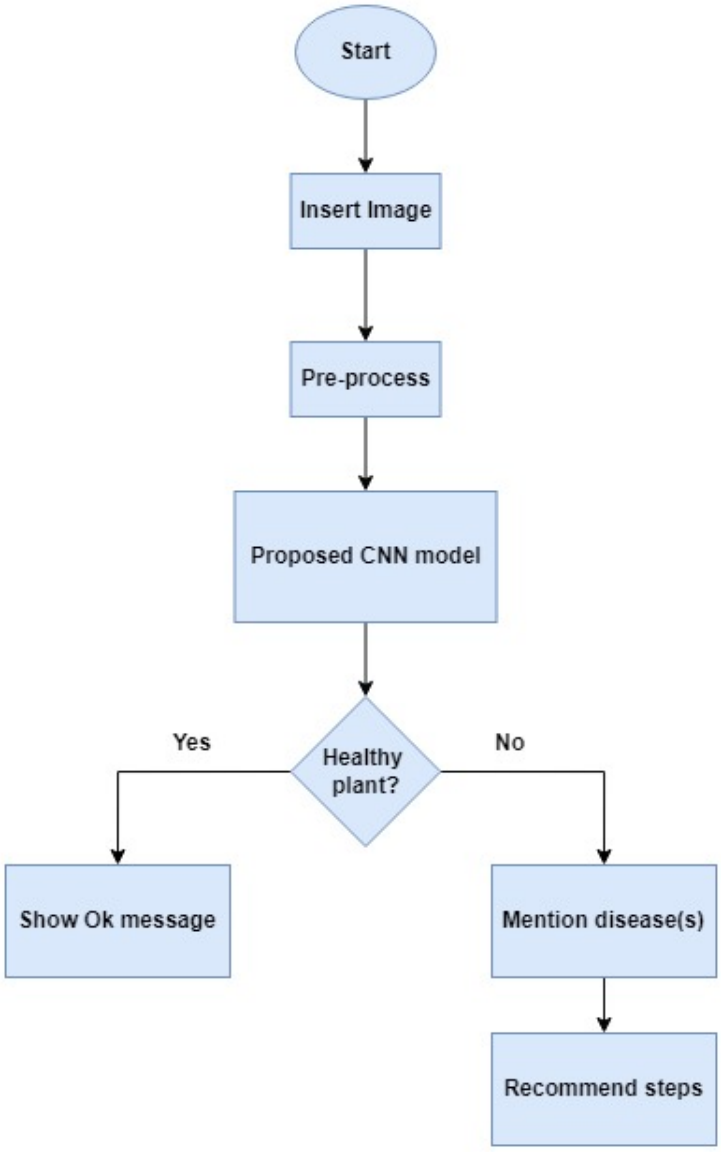


Figure 3.2: Workflow of the android application

Chapter 4

Conclusion

Most of the people in our country depend on agriculture. So it's very important to find the disease as early as possible with modern technology. Through this research, we aim to implement a Deep learning model for detecting various kinds of wheat diseases. Creating an automated plant disease detection system with high precision and low resource requirements would save farmers time and money. In comparison to existing pre-trained models, we anticipate gaining greater illness detection accuracy with our suggested CNN model. To sum up we will create an app which will take images and detect if any illness exists and recommend the farmers of necessary actions to be performed.

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